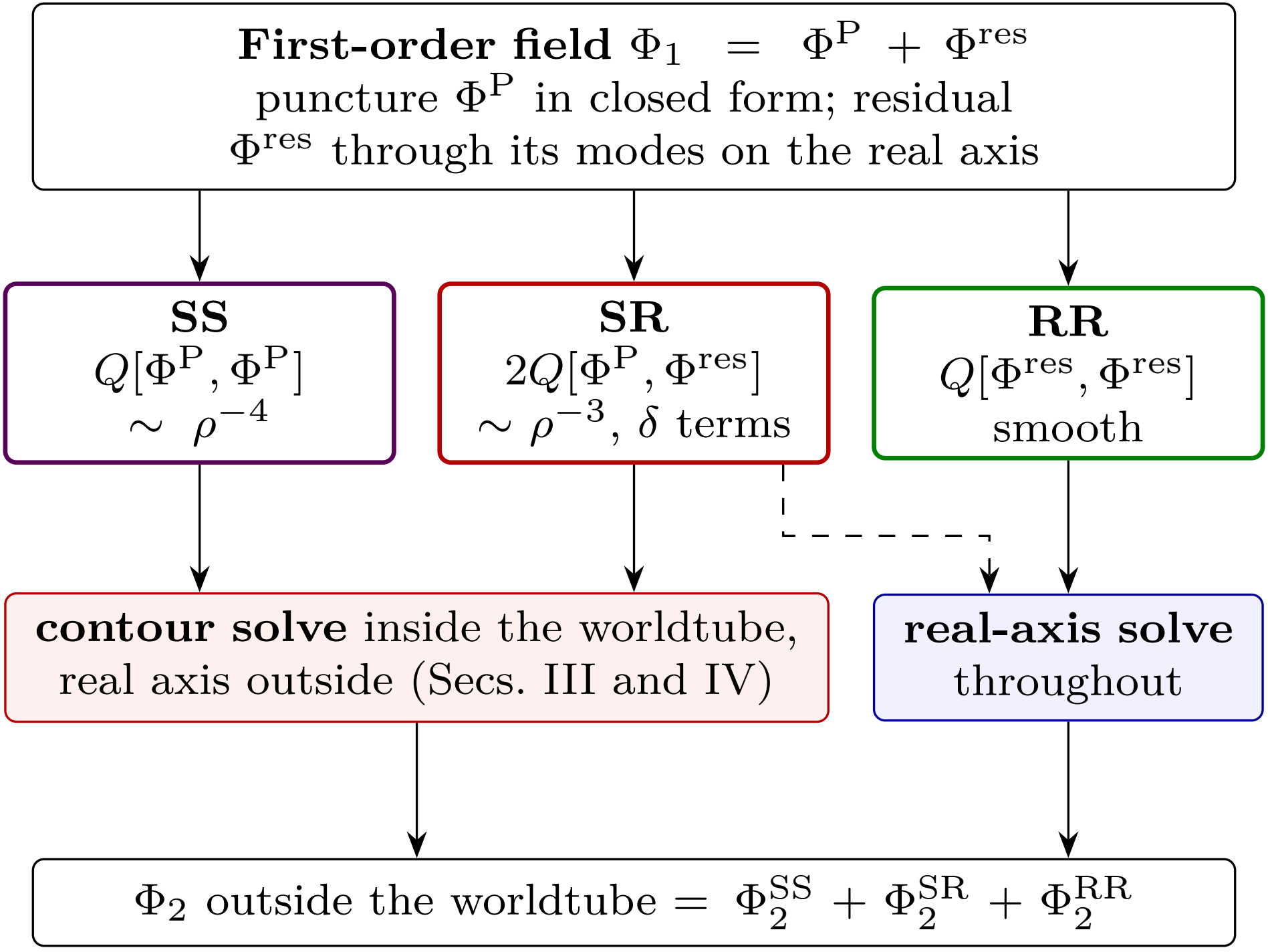

First-order field $\Phi_1 = \Phi^{\rm P} + \Phi^{\rm res}$
puncture $\Phi^{\rm P}$ in closed form; residual
$\Phi^{\rm res}$ through its modes on the real axis
SS
$Q[\Phi^{\rm P}, \Phi^{\rm P}]$
$\sim \rho^{-4}$
SR
$2Q[\Phi^{\rm P}, \Phi^{\rm res}]$
$\sim \rho^{-3}$, δ terms
RR
$Q[\Phi^{\rm res}, \Phi^{\rm res}]$
smooth
contour solve inside the worldtube,
real axis outside (Secs. III and IV)
real-axis solve
throughout
Φ_2 outside the worldtube $= \Phi_2^{\rm SS} + \Phi_2^{\rm SR} + \Phi_2^{\rm RR}$